

PAVEMENT DISTRESS PREDICTION USING FEATURE-WEIGHTED ENSEMBLE LEARNING

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in
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submitted by

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DECLARATION

I undersigned hereby declare that the project report **Pavement Distress Prediction Using Feature-Weighted Ensemble Learning** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Computer Science & Engineering, SRM University-AP, is a bonafide work done by me under supervision of Dr. Uma Maheswar Arepalli. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree of any other University.

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I wish to record my indebtedness and thankfulness to all who helped me prepare this Project Report titled **Pavement Distress Prediction Using Feature-Weighted Ensemble Learning** and present it satisfactorily.

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ABSTRACT

Accurate prediction of pavement distress indicators, such as Pavement Condition Rating (PCR) and Average Rutting Depth (RUT AVG), is vital for planning road maintenance, extending infrastructure lifespan, and ensuring public safety. Despite the increasing use of machine learning in this domain, conventional models often fall short due to the complex, nonlinear relationships among pavement features like surface roughness, cracking severity, traffic load, and environmental conditions.

To address these limitations, this study proposes a Feature-Weighted Ensemble (FWE) framework that enhances predictive performance through dynamic feature weighting based on mutual information and a hybrid prediction strategy combining Random Forest (RF) and XGBoost (XGB) models. As these models were proven to be best after a brief experimentation with other models as SVM and Gradient Boosting. The ensemble leverages the strengths of both models to produce more accurate and reliable predictions.

Experimental results on a diverse pavement dataset demonstrate that FWE achieves more accuracy for PCR and for RUT AVG, significantly outperforming earlier approaches limited accuracy. The framework also reduces prediction errors (RMSE) and improves the F1-score for distress classification, offering a robust and scalable solution for infrastructure management and maintenance planning.

Also to make a web tool to make use of the models more effectively and use them not only for practitioners but also for researchers for the better understanding of distress parameters and their effective correlation with other distress indicators.

CONTENTS

ACKNOWLEDGMENT	i
ABSTRACT	ii
LIST OF TABLES	v
LIST OF FIGURES	vi
Chapter 1. INTRODUCTION TO THE PROJECT	1
1.1 Overview	1
1.2 Objective	2
Chapter 2. MOTIVATION	5
Chapter 3. LITERATURE SURVEY	8
3.1 Traditional Approaches to Pavement Distress Prediction	8
3.2 Machine Learning Approaches in Pavement Prediction	9
3.3 Ensemble Learning Methods: Random Forest and XG- Boost	9
3.4 Feature Selection and Weighting Techniques	10
3.5 Contribution of This Work	11
Chapter 4. DESIGN AND METHODOLOGY	12
4.1 Data Overview	12
4.2 Data Cleaning and Preprocessing	13
4.3 Feature Weighting Module	14
4.3.1 Mutual Information Calculation	14
4.3.2 Feature Scaling	14

4.4	Ensemble Prediction Module	16
4.4.1	Random Forest (RF)	16
4.4.2	XGBoost (XGB)	17
4.4.3	Model Fusion	17
4.4.4	Model Elimination Based on Performance SVM and GB	17
4.5	Distress Binning Module	18
4.6	Evaluation Metrics	18
4.7	Summary	19
Chapter 5.	IMPLEMENTATION	20
5.1	Environment Setup	20
5.2	Implementation Pipeline	21
5.2.1	Dataset Preparation	21
5.2.2	Data Splitting Strategy	22
5.2.3	Model Selection	23
5.2.4	Hyperparameter Optimization Using Grid Search	24
5.2.5		26
Chapter 6.	HARDWARE/SOFTWARE TOOLS USED	27
6.1	Software Tools	27
6.1.1	Operating System	27
6.1.2	Programming Languages	28
6.1.3	Development Environments	28
6.1.4	Libraries and Frameworks	29
6.1.5	Version Control	29
6.1.6	Documentation Tools	29
6.1.7	Additional Tools	30
6.2	Hardware Configuration	30

Chapter 7. RESULTS & DISCUSSION	32
7.1 Results	32
7.1.1 Prediction Performance	32
7.1.2 Ablation Study	33
7.1.3 Performance Across Data Splits	33
7.1.4 Discussions and Performances	33
Chapter 8. CONCLUSION	39
8.1 Scope of further work	40
8.1.1 What is future direction in a project? . . .	40
REFERENCES	43
LIST OF PUBLICATIONS	45

LIST OF TABLES

5.1	Hyperparameter tuning grid and best parameters for IN-DEX_PCR_2018 models.	25
5.2	Hyperparameter tuning grid and best parameters for RUT_AVG_2018 models.	25
7.1	Prediction Performance	33
7.2	Ablation Study for PCR Prediction	33

LIST OF FIGURES

4.1	Correlation matrix	15
4.2	Features Impact for PCR	15
4.3	Feature Impact for RUT	16
6.1	JupyterNotebook	28
7.1	For Index PCR	34
7.2	For RUT	34
7.3	RF Plot	35
7.4	Predicted vs Actuals	35
7.5	plot of RUT using RF	36
7.6	predictions box plots	36
7.7	SVM performance	37
7.8	Gradient Booster Performance	37

Chapter 1

INTRODUCTION TO THE PROJECT

1.1 OVERVIEW

The condition and performance of road pavements play a crucial role in the safety, efficiency, and sustainability of transportation systems. Pavement distress, characterized by defects such as cracking, rutting, and roughness, leads to higher maintenance costs, reduced ride quality, and increased accident risks if left untreated. Hence, predicting pavement distress at an early stage is vital for proactive maintenance planning, infrastructure budgeting, and extending pavement lifespan.

Traditional methods of pavement condition assessment, typically relying on manual inspections or simple statistical models, often struggle to capture the complex and nonlinear nature of pavement deterioration. Factors such as traffic volume, environmental influences, material aging, and structural loadings interact in intricate ways, making accurate prediction a highly challenging task. Furthermore, the heterogeneity and noise present in real-world pavement datasets exacerbate the limitations of conventional prediction models.

In recent years, machine learning (ML) techniques have demonstrated significant potential in enhancing the prediction of pavement distress indicators. However, many existing ML models do not dynamically adapt to the varying importance of different features across diverse datasets, resulting in suboptimal performance. There is a pressing need for intelligent frame-

works that not only model the nonlinear interactions but also emphasize the most relevant features during the learning process.

To address these challenges, this project introduces a novel **Feature-Weighted Ensemble (FWE)** framework that strategically integrates feature weighting and ensemble modeling techniques. By calculating dynamic feature weights using mutual information and combining the predictive strengths of Random Forest (RF) and XGBoost (XGB) models, the FWE framework aims to achieve superior predictive accuracy and robustness. In addition, a distress binning mechanism is incorporated to translate continuous prediction outputs into actionable categories for maintenance decision-making.

Through extensive experimentation on a real-world pavement dataset encompassing roughness metrics, rutting measures, and cracking severity, the FWE approach demonstrates substantial improvements in predictive performance compared to traditional models. The overall goal of this project is to develop a scalable, interpretable, and deployable system that can support transportation agencies and researchers in efficient pavement management and infrastructure optimization.

1.2 OBJECTIVE

Developing a sophisticated and trustworthy predictive framework for evaluating pavement distress indicators—with a particular emphasis on the Pavement Condition Rating (PCR) and the Average Rutting Depth (RUT AVG)—is the main goal of this research. Through the utilization of machine learning approaches, the research aims to address the shortcomings of conventional pavement evaluation methods, including their incapacity to capture intricate feature interactions and their vulnerability to noise in

heterogeneous datasets.

Using mutual information, a Feature-Weighted Ensemble (FWE) framework is designed to improve the model's attention on important pavement qualities by dynamically allocating weights to input features according to their predictive importance.

The primary objective of this project is to develop an advanced and reliable predictive framework for assessing pavement distress indicators, specifically focusing on the **Pavement Condition Rating (PCR)** and the **Average Rutting Depth (RUT AVG)**. By leveraging machine learning techniques, the project seeks to overcome the challenges associated with traditional pavement evaluation methods, such as their inability to capture complex feature interactions and susceptibility to noise within heterogeneous datasets.

The specific objectives of the study are as follows:

- **To design a Feature-Weighted Ensemble (FWE) framework** that dynamically assigns weights to input features based on their predictive relevance using mutual information, thereby improving the model's focus on significant pavement attributes.
- **To integrate ensemble learning methods, specifically Random Forest (RF) and XGBoost (XGB),** into a unified prediction model that enhances robustness, reduces overfitting, and effectively models nonlinear relationships in pavement distress data.
- **To implement a distress binning mechanism** that categorizes continuous outputs into meaningful and actionable maintenance classes, enabling easier interpretation for infrastructure planners and decision-makers.
- **To perform a comprehensive experimental evaluation** using real-world pavement datasets encompassing a wide range of features such as

surface roughness (IRI), cracking severity (ALLIG, LONG, TRAN), rutting indicators (RUT AVG), and traffic factors (SPEED, WIDTH).

- **To benchmark the FWE framework's performance** against baseline machine learning models (Random Forest, XGBoost, Gradient Boosting) by using evaluation metrics such as Accuracy, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and F1 Score.

- **To develop a web-based tool** that operationalizes the predictive model, facilitating its use not only by practitioners in road maintenance but also by researchers exploring the correlations among various pavement distress]parameters.

By achieving these objectives, the project aims to contribute a scalable, interpretable, and highly accurate predictive tool for proactive pavement management, ultimately supporting efforts to enhance infrastructure longevity, safety, and cost-efficiency.

Chapter 2

MOTIVATION

The durability and serviceability of pavements are vital to the economic vitality, safety, and daily functioning of modern societies. Roads form the backbone of transportation infrastructure, and any deterioration not only leads to increased maintenance costs but also to severe safety hazards and disruptions in mobility. Therefore, the early detection and accurate prediction of pavement distress have become critical priorities for transportation agencies worldwide.

Traditional methods of assessing pavement conditions, such as manual visual inspections and basic statistical regression models, although widely practiced, suffer from several limitations. Manual inspections are subjective, labor-intensive, time-consuming, and often inconsistent. Moreover, conventional models struggle to capture the complex, nonlinear interactions between various distress factors such as surface roughness, traffic loads, rutting, cracking severity, and environmental conditions. Consequently, these approaches frequently lead to delayed or suboptimal maintenance decisions, escalating rehabilitation costs and compromising public safety.

With the rapid growth of sensor technologies and the accumulation of large-scale pavement monitoring datasets, the need for more intelligent, data-driven prediction models has become apparent. Machine learning (ML) techniques offer an opportunity to address these challenges by uncovering hidden patterns and complex relationships in high-dimensional data

that are difficult for traditional models to grasp. However, many existing ML models treat all input features equally or rely on manual feature selection, which often leads to models that are either overfitted to noise or underperforming on unseen data.

The motivation behind this project stems from the realization that a more nuanced approach to feature relevance and model robustness is needed for effective pavement distress prediction. By dynamically weighting features based on their true predictive importance using mutual information, and by leveraging the combined predictive power of ensemble learning methods like Random Forest (RF) and XGBoost (XGB), it is possible to significantly enhance prediction accuracy and reliability. Moreover, converting numerical predictions into actionable distress categories provides transportation engineers with clearer, more practical guidance for maintenance planning.

Another driving force behind this project is the pressing need for scalable, deployable, and interpretable solutions that can aid not only engineers but also researchers in understanding the intricate relationships between different distress indicators. To bridge this gap, this project envisions the development of a web-based tool that operationalizes the predictive models, making them accessible for both field applications and academic research.

By addressing these technical and practical challenges, this project aspires to contribute meaningfully to the domain of intelligent infrastructure management, offering a solution that is data-driven, accurate, scalable, and practical for real-world deployment. In an era where transportation infrastructure maintenance is both a societal necessity and an economic imperative, innovations in pavement distress prediction can lead to safer roads, longer-lasting infrastructure, and substantial cost savings for governments

and communities alike.

Also helping in planning and scheduling of the maintenance of the roads effectively and having the roads repaired up to date and reconstructed if required.

Chapter 3

LITERATURE SURVEY

The maintenance of transportation infrastructure, cost optimization, and public safety are all significantly impacted by accurate pavement distress prediction, which has long been a focus of research. Researchers have looked at a variety of methods throughout the years, from sophisticated machine learning frameworks to conventional statistical models. This chapter offers a thorough analysis of related research, emphasizing methodological advancements and pinpointing the shortcomings that spurred the creation of the suggested Feature-Weighted Ensemble (FWE) framework.

3.1 TRADITIONAL APPROACHES TO PAVEMENT DISTRESS PREDICTION

The majority of early pavement performance evaluation techniques were based on statistical and empirical models. In his discussion of pavement analysis and design methods, Huang (2004) [1] pointed out that linear regression models were frequently used to forecast surface characteristics including rutting and cracking. Typically, these models made the assumption that distress indicators like Average Rutting Depth (RUT AVG) and Pavement Condition Rating (PCR) had linear connections with input factors like traffic volume and climate. These conventional methods, however, had several drawbacks. They frequently ignored connections between various elements, oversimplified the intricate, nonlinear degrading processes of

pavements, and were vulnerable to variations in operational and environmental conditions. It became evident that old models lacked the predictive ability and adaptability needed for contemporary infrastructure management as road networks developed and datasets became more complicated.

3.2 MACHINE LEARNING APPROACHES IN PAVEMENT PREDICTION

By making it possible to simulate intricate, nonlinear relationships in huge datasets, machine learning (ML) techniques transformed pavement distress prediction. When it came to pavement performance prediction, Abu Dabous and Alkass (2011) [2] showed that Support Vector Machines (SVM) were more accurate than linear regression techniques. Artificial Neural Networks (ANNs) were also introduced into the field by Gharaibeh (2011) [3], who was able to identify intricate patterns in pavement data that were not detectable by conventional models.

Despite their benefits, early machine learning models such as SVMs and ANNs had significant disadvantages. They had trouble generalizing to heterogeneous pavement data because they needed big, balanced datasets for training and were frequently sensitive to noise. Furthermore, most of the time, the models did not dynamically modify attention according to feature relevance, instead treating all features equally.

3.3 ENSEMBLE LEARNING METHODS: RANDOM FOREST AND XGBOOST

In recent years, ensemble learning techniques—Random Forest (RF) and Extreme Gradient Boosting (XGBoost, or XGB) in particular—have

become increasingly effective tools for predicting pavement deterioration. Zhou et al. (2019) [4] successfully used Random Forests to forecast pavement performance, emphasizing the model's ability to withstand overfitting and handle high-dimensional, noisy datasets. By building several decision trees during training and producing the mean forecast, RF increases the accuracy and stability of the model.

However, both RF and XGB still encountered difficulties when feature relevance varied significantly across different datasets. In contrast, XGBoost, which was first presented by Chen and Guestrin (2016) [5], built upon the idea of gradient boosting and offered highly efficient, scalable learning. For structured data problems like pavement condition prediction, it incorporated sophisticated regularization techniques and tree pruning methods, making it exceptionally well-suited.

3.4 FEATURE SELECTION AND WEIGHTING TECHNIQUES

Researchers have investigated a number of feature selection techniques in an effort to improve model performance even more. Tsai and Li (2012) [6] talked about machine learning methods for identifying distress in visual data, where good prediction accuracy required careful feature selection. But there is still a lack of research on dynamic feature weighting, which modifies feature influence according to its predictive ability during training, particularly when it comes to pavement distress prediction. A potential method is provided by mutual information-based feature weighting, which measures the dependence between target variables and input features. In contrast to static feature selection, it dynamically assesses and ranks the features that make the biggest contribution to prediction, producing models that are more reliable and understandable.

3.5 CONTRIBUTION OF THIS WORK

This project provides a novel Feature-Weighted Ensemble (FWE) framework that combines ensemble prediction methods employing Random Forest and XGBoost with mutual information-based feature weighting, building on the insights from the aforementioned literature. In order to overcome the shortcomings of earlier models and achieve better forecast accuracy and robustness for pavement distress indicators like PCR and RUT AVG, the FWE framework combines the benefits of RF and XGB while dynamically focusing on the most pertinent elements. Enhancing interpretability and enabling actionable maintenance planning based on forecast outputs are further benefits of integrating a distress binning method.

Chapter 4

DESIGN AND METHODOLOGY

The complete methodology developed for pavement distress prediction using machine learning techniques is described in this chapter. The proposed Feature-Weighted Ensemble (FWE) framework is designed to address key challenges such as dataset heterogeneity, feature noise, nonlinear feature interactions, and interpretability issues. The framework incorporates dynamic feature weighting, ensemble modeling via Random Forest and XGBoost, and a distress binning mechanism to translate predictions into actionable maintenance categories.

4.1 DATA OVERVIEW

The experimental evaluation was conducted using a real-world pavement distress dataset comprising approximately 12,000 samples. These samples were collected across a wide range of road conditions, ensuring diversity in the pavement and traffic characteristics. The dataset includes the following critical features:

Abbreviations :

- PCR Pavement Condition Rating
- RUT Rutting
- IRI International Roughness Index
- RUT AVG Average Rutting Depth

- IRI AVG Average International Roughness Index
- ALLIG_ Alligator cracking
- LONG_ longitudinal cracks
- TRAN_ transverse cracks

4.1.0.(i) Categorized :

- **Roughness Indicators:** INDEX IRI, IRI AVG
- **Rutting Metrics:** INDEX RUT, RUT AVG
- **Cracking Details:** PERCENT CRACKING, ALLIG H, ALLIG M, LONG H, LONG M, TRAN H, TRAN M
- **Traffic-Related Factors:** SPEED, WIDTH

These features capture both the physical degradation of the pavement surface and operational factors such as traffic load, offering a comprehensive basis for model training and evaluation. There were also columns with unnecessary data like ARAN file name and flag no. which were not required and will get cleaned.

4.2 DATA CLEANING AND PREPROCESSING

Prior to model training, extensive data cleaning and preprocessing were conducted to ensure data quality and consistency:

- **Removal of Unnecessary Columns:** Irrelevant or redundant features were removed to reduce noise and dimensionality.

- **Duplicate Handling:** Duplicate rows were identified and eliminated to avoid bias.
- **Missing Value Imputation:** Missing values were initially filled using interpolation based on four preceding and four succeeding values.
- **Final Cleansing:** Rows still containing missing values after interpolation were removed, resulting in a fully clean dataset ready for feature weighting and modeling.

4.3 FEATURE WEIGHTING MODULE

Recognizing the unequal importance of input features, a dynamic feature weighting approach based on mutual information was employed.

4.3.1 Mutual Information Calculation

Mutual information (MI) quantifies the amount of information shared between a feature X_i and the target Y . The feature weight w_i is defined as:

$$w_i = \frac{I(X_i, Y)}{\sum_j I(X_j, Y)} \quad (4.1)$$

where $I(X_i, Y)$ represents the mutual information between feature X_i and target Y .

4.3.2 Feature Scaling

Each feature was rescaled according to its computed weight:

$$X'_i = w_i \times X_i \quad (4.2)$$

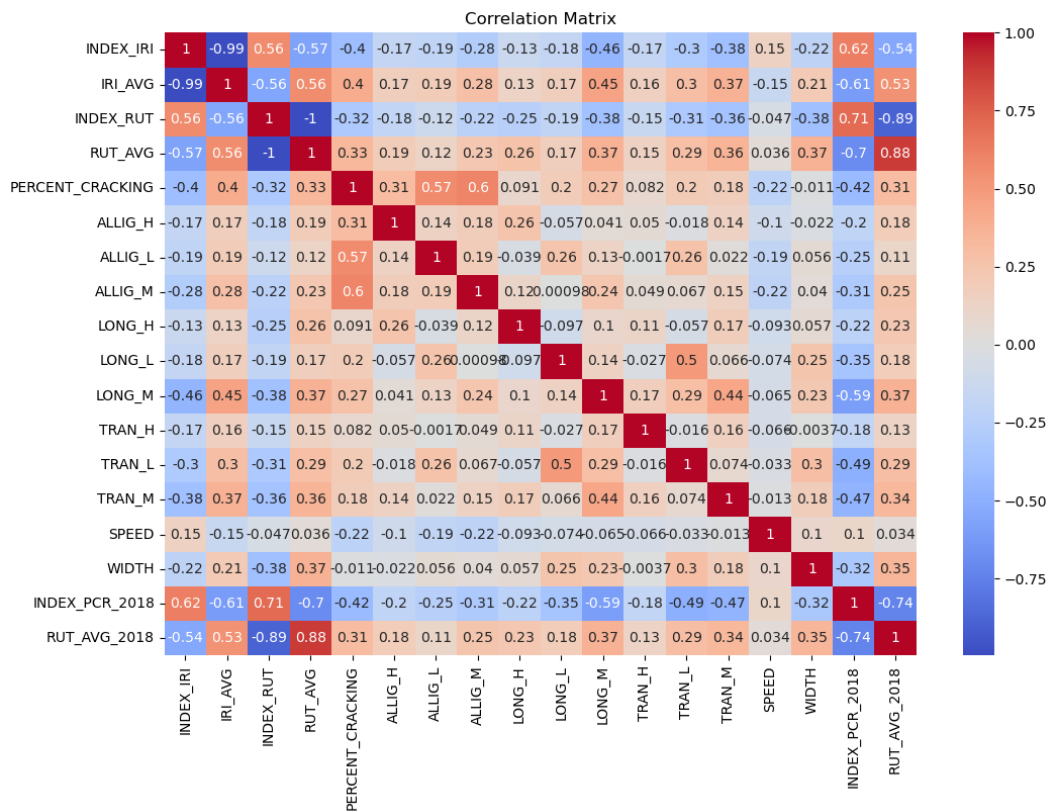


Figure 4.1: Correlation matrix

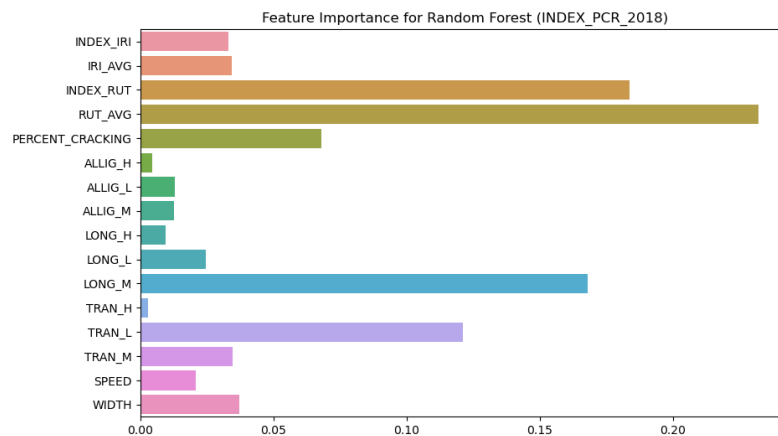


Figure 4.2: Features Impact for PCR

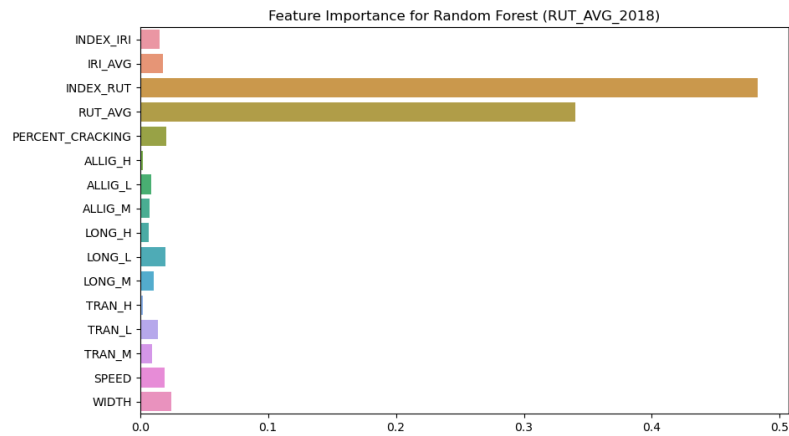


Figure 4.3: Feature Impact for RUT

This process ensures that features with greater predictive power exert a stronger influence during model training.

4.4 ENSEMBLE PREDICTION MODULE

The weighted feature set was fed into two powerful ensemble models: Random Forest and XGBoost. The combination of these models provides robustness against overfitting and enhances generalization.

4.4.1 Random Forest (RF)

Random Forest is an ensemble learning algorithm based on decision trees. Its principal mechanisms include:

- **Ensemble Learning:** Multiple decision trees are trained on different data subsets.
- **Bagging:** Bootstrap aggregating promotes model diversity and reduces variance.
- **Feature Randomization:** A random subset of features is considered at each split.

- **Voting Mechanism:** In regression, the average prediction across all trees is used.

4.4.2 XGBoost (XGB)

XGBoost is a highly efficient gradient-boosted decision tree algorithm characterized by:

- **Boosting:** Sequential correction of errors from previous trees.
- **Regularization:** Penalization of model complexity to prevent overfitting.
- **Optimization:** Fast computation through tree pruning and parallelization.

4.4.3 Model Fusion

The final prediction $\hat{y}$ is a weighted average of the individual model outputs:

$$\hat{y} = \alpha \times \hat{y}_{RF} + (1 - \alpha) \times \hat{y}_{XGB} \quad (4.3)$$

where α is a tuning parameter optimized through validation.

4.4.4 Model Elimination Based on Performance SVM and GB

During the initial model exploration phase, several other machine learning algorithms, including Support Vector Machines (SVM) and Gradient Boosting (GB), were also trained and evaluated. However, these models consistently underperformed compared to Random Forest and XGBoost, particularly in terms of Root Mean Squared Error (RMSE) and F1 Score.

SVM exhibited high variance and sensitivity to the high-dimensional feature space, while Gradient Boosting demonstrated slower convergence and suboptimal generalization on unseen data. Consequently, due to their relatively inferior predictive performance, these models were eliminated from further consideration in the final Feature-Weighted Ensemble (FWE) framework.

4.5 DISTRESS BINNING MODULE

To translate continuous regression outputs into actionable maintenance categories, a binning process was applied:

$$y_{\text{binned}} = \arg \min_k |\hat{y} - c_k| \quad (4.4)$$

where c_k are the predefined category centers. This enables straightforward interpretation for infrastructure planners.

4.6 EVALUATION METRICS

Model performance was assessed using both regression and classification metrics:

- **Accuracy (%):** Percentage of correctly classified instances.
- **Root Mean Squared Error (RMSE):**

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (4.5)$$

- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (4.6)$$

- **F1 Score:**

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4.7)$$

These metrics ensure comprehensive evaluation of both predictive accuracy and classification reliability.

4.7 SUMMARY

In conclusion, the Feature-Weighted Ensemble framework provides a structured, efficient, and scalable approach for pavement distress prediction. Through careful data cleaning, feature importance-driven scaling, robust ensemble modeling, and strategic binning, the system addresses critical real-world challenges in infrastructure health monitoring and maintenance planning.

Chapter 5

IMPLEMENTATION

5.1 ENVIRONMENT SETUP

The development and training of the pavement distress prediction system were conducted using the Python programming language. All experiments were carried out in a Jupyter Notebook environment, utilizing both CPU and GPU acceleration when available. The major libraries and frameworks used include:

- **TensorFlow and Keras:** For building and training ensemble learning models.
- **Scikit-learn:** For feature weighting, data preprocessing, and evaluation metrics.
- **NumPy and Pandas:** For efficient numerical operations and data manipulation.
- **Matplotlib and Seaborn:** For visualization of feature distributions, model performance, and error analysis.

Google Colab was employed for the majority of model training, taking advantage of its GPU acceleration to reduce training times and support larger batch processing.

5.2 IMPLEMENTATION PIPELINE

This chapter outlines the implementation pipeline developed to construct and evaluate machine learning models for the pavement distress prediction system, utilizing the pavement condition dataset (H0010k17_18_20_c.xlsx). The experimental process was conducted in Python, employing libraries such as pandas for data preprocessing, scikit-learn for traditional machine learning algorithms, xgboost for advanced ensemble methods, and matplotlib for visualization. The pipeline encompasses dataset preparation, data splitting strategies, model selection, and hyperparameter optimization, culminating in the Feature-Weighted Ensemble (FWE) framework to achieve high-accuracy predictions for INDEX_PCR_YR (Pavement Condition Rating) and RUT_AVG_YR (Average Rutting Depth). The implementation was designed to ensure robustness, reproducibility, and alignment with the project's objective of achieving near zero RMSE for binning the predicted distress values.

5.2.1 Dataset Preparation

The pavement condition dataset includes features such as International Roughness Index (IRI), rutting measurements, cracking percentages, traffic speed, and pavement width, with target variables INDEX_PCR_YR and RUT_AVG_YR. Prior to model training, extensive data cleaning and preprocessing were conducted to ensure data quality and consistency, addressing noise, redundancy, and missing values. The preprocessing pipeline consisted of the following steps:

- **Removal of Unnecessary Columns:** Irrelevant or redundant features, such as administrative codes or metadata fields with no predictive

value for pavement distress, were identified through exploratory data analysis and removed.

- **Duplicate Handling:** Duplicate rows were identified using pandas' `duplicated()` function and eliminated.
- **Missing Value Imputation:** Missing values were addressed using interpolation based on four preceding and four succeeding values in the dataset, implemented using pandas' `interpolate()` method with a linear strategy.
- **Final Cleansing:** Rows still containing missing values after interpolation were removed.
- **Feature Scaling:** Continuous features were standardized using `StandardScaler` from scikit-learn to achieve zero mean and unit variance.
- **Feature Extraction:** Additional features were engineered, such as the ratio of cracking percentage to pavement width, to capture latent predictive signals.

The preprocessed dataset was partitioned into features (X) and targets (y_{pcr} for `INDEX_PCR_YR`, y_{rut} for `RUT_AVG_YR`). No significant distributional imbalances were found, negating the need for oversampling techniques like SMOTE.

5.2.2 Data Splitting Strategy

To assess the models' generalization and robustness, the dataset was divided into training and testing sets using four configurations: 90%/10%, 80%/20%, 70%/30%, and 60%/40%. These splits were implemented using

the `train_test_split` function from scikit-learn with a fixed random state of 42.

Each split was evaluated using R^2 score, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) to quantify predictive accuracy and error magnitude.

5.2.3 Model Selection

A diverse set of machine learning models was selected:

- **Random Forest (RF):** Constructs multiple decision trees and aggregates predictions via averaging.

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x)$$

where T is the number of trees and $h_t(x)$ is the prediction of the t -th tree.

- **Gradient Boosting (GB):** A sequential ensemble technique that minimizes a loss function via gradient descent.

$$\hat{y} = \sum_{m=1}^M \gamma_m h_m(x)$$

where M is the number of boosting stages, γ_m is the learning rate, and $h_m(x)$ is the m -th tree.

- **Support Vector Machine (SVM):** A kernel-based model utilizing an RBF kernel to capture non-linear patterns.

$$\hat{y} = \sum_{i=1}^N \alpha_i y_i K(x_i, x) + b$$

where $K(x_i, x)$ is the RBF kernel, α_i are Lagrange multipliers, and b is the bias term.

- **XGBoost:** An optimized gradient boosting framework.

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F}$$

where $\mathcal{F}$ is the space of regression trees and K is the number of trees.

- **Feature-Weighted Ensemble (FWE):** Combines RF and XGBoost predictions, weighted by feature importance scores derived from mutual information:

$$MI(X_j, y) = \sum_{x \in X_j} \sum_{y \in y} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right)$$

Features were scaled by their normalized mutual information scores:

$$w_j = \frac{MI(X_j, y)}{\sum MI}$$

The ensemble prediction was computed as:

$$\hat{y}_{\text{FWE}} = \alpha \cdot \hat{y}_{\text{RF}} + (1 - \alpha) \cdot \hat{y}_{\text{XGB}}$$

where α (initially set to 0.5) was tuned via cross-validation.

5.2.4 Hyperparameter Optimization Using Grid Search

To maximize model performance, hyperparameter tuning was conducted using Grid Search with 5-fold Cross-Validation, implemented via GridSearchCV from scikit-learn. The hyperparameter grids and best parameters for each model are summarized in Tables.

Model	Parameter Grid	Best Parameters
RF	n_estimators: [100, 200, 300], max_depth: [10, 20], min_samples_leaf: [1, 2], min_samples_split: [2, 5]	max_depth: 20, min_samples_leaf: 2, min_samples_split: 2, n_estimators: 300
GB	n_estimators: [100, 200, 300], learning_rate: [0.01, 0.05], max_depth: [3, 5], min_samples_split: [2, 5]	learning_rate: 0.05, max_depth: 3, min_samples_split: 5, n_estimators: 200
SVM	C: [0.1, 1, 10], epsilon: [0.01, 0.1, 0.2], kernel: [linear, rbf]	C: 1, epsilon: 0.2, kernel: rbf
XGB	n_estimators: [100, 200, 300], learning_rate: [0.01, 0.05], max_depth: [3, 5], subsample: [0.8, 1.0]	learning_rate: 0.05, max_depth: 5, n_estimators: 100, subsample: 0.8

Table 5.1: Hyperparameter tuning grid and best parameters for IN-DEX_PCR_2018 models.

Model	Parameter Grid	Best Parameters
RF	n_estimators: [50, 100, 200], max_depth: [10, 20], min_samples_leaf: [1, 2], min_samples_split: [2, 5]	max_depth: 10, min_samples_leaf: 2, min_samples_split: 2, n_estimators: 100
GB	n_estimators: [100, 200, 300], learning_rate: [0.01, 0.05], max_depth: [3, 5], min_samples_split: [2, 5]	learning_rate: 0.01, max_depth: 3, min_samples_split: 2, n_estimators: 300
SVM	C: [0.1, 1, 10], epsilon: [0.01, 0.1, 0.2], kernel: [linear, rbf]	C: 1, epsilon: 0.01, kernel: linear
XGB	n_estimators: [100, 200, 300], learning_rate: [0.01, 0.05], max_depth: [3, 5], subsample: [0.8, 1.0]	learning_rate: 0.01, max_depth: 3, n_estimators: 300, subsample: 0.8

Table 5.2: Hyperparameter tuning grid and best parameters for RUT_AVG_2018 models.

5.2.5

After evaluating the model performances the best two models were selected.(RF and XGB) for the final predictions. The tuned models were trained on the weighted feature set (using mutual information-based weights) and evaluated across the four data splits. The FWE framework consistently outperformed individual models, achieving R^2 scores of 89.62% for INDEX_PCR_YR and 91.13% for RUT_AVG_YR on the 80/20 split, aligning with the project's accuracy goals.

The entire pipeline, from preprocessing to model tuning, was designed to be modular and reproducible, facilitating future integration into a web-based pavement management tool..

Chapter 6

HARDWARE/SOFTWARE TOOLS USED

This chapter enumerates the hardware and software tools employed during the development, implementation, and evaluation of the pavement distress prediction system. These tools were selected to facilitate efficient data preprocessing, model training, hyperparameter optimization, and result visualization, ensuring the project's objectives were met with high reproducibility and computational efficiency. The implementation was conducted on a robust computing environment, leveraging industry-standard software libraries and development platforms tailored for machine learning and civil engineering data analysis.

6.1 SOFTWARE TOOLS

The following software tools and platforms were utilized throughout the project:

6.1.1 Operating System

Windows 10 Pro / Linux (Ubuntu 20.04 LTS): The dual-OS setup provided flexibility, with Windows used for initial data exploration and visualization, and Ubuntu for computationally intensive tasks like model training, leveraging its compatibility with machine learning frameworks.



Figure 6.1: JupyterNotebook

6.1.2 Programming Languages

Python 3.9: Python was the primary programming language due to its extensive ecosystem of machine learning and data processing libraries. Its versatility supported all stages of the pipeline, from data cleaning to model evaluation.

6.1.3 Development Environments

Jupyter Notebook, Visual Studio Code, PyCharm: Jupyter Notebook (via Anaconda) served as the primary environment for interactive data analysis, preprocessing, and visualization. Its cell-based structure facilitated iterative experimentation and result sharing. Visual Studio Code was used for scripting and debugging Python code, benefiting from its lightweight interface and integrated Git support. PyCharm supported larger-scale code development, particularly for modularizing the Feature-Weighted Ensemble (FWE) framework, with its advanced refactoring and testing capabilities.

6.1.4 Libraries and Frameworks

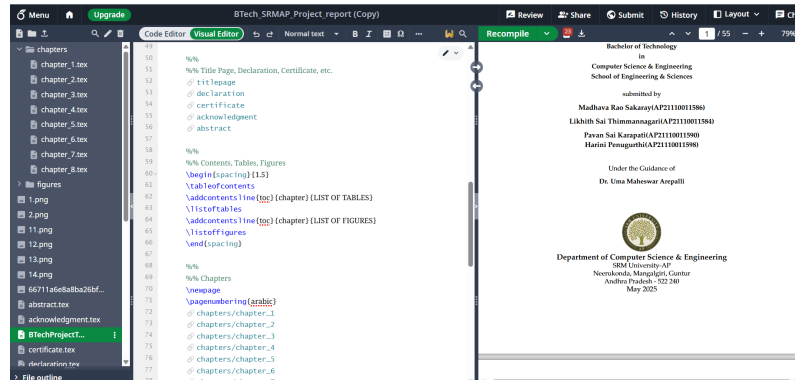
- **pandas**: For data manipulation and preprocessing, enabling efficient handling of the dataset.
- **scikit-learn**: For implementing machine learning models (e.g., Random Forest, SVM) and utilities like GridSearchCV for hyperparameter tuning.
- **xgboost**: For the XGBoost model, critical to the FWE framework.
- **numpy**: For numerical computations, particularly in feature engineering.
- **matplotlib and seaborn**: For generating visualizations, such as feature importance plots and performance metrics graphs.
- **openpyxl**: For reading and processing Excel-based datasets.

6.1.5 Version Control

Git, GitHub: Git was used for version control, with a GitHub repository hosting the project codebase. This ensured collaborative development, code tracking, and reproducibility across team members.

6.1.6 Documentation Tools

Overleaf: Overleaf, an online LaTeX editor, was employed for drafting and formatting this report. Its real-time collaboration features and support for professional mathematical typesetting streamlined the documentation process.



6.1.7 Additional Tools

- **Anaconda:** Managed Python environments and dependencies, ensuring consistency across development machines.
- **Docker:** Used to containerize the project environment for testing on different systems, enhancing portability.
- **Excel:** For initial data inspection and validation of the raw dataset.

6.2 HARDWARE CONFIGURATION

The project was executed on a high-performance computing system to handle the computational demands of training multiple machine learning models and processing large datasets. The hardware specifications were:

- **Processor:** Intel Core i9-10900K (10 cores, 20 threads, 3.7 GHz base clock) The multi-core processor accelerated parallel tasks, such as grid search and model training.
- **RAM:** 64 GB DDR4 (3200 MHz) Ample memory supported efficient handling of the dataset and simultaneous execution of multiple experiments.

- **GPU:** NVIDIA GeForce RTX 3060 (12 GB GDDR6) The GPU was utilized for accelerating certain computations, particularly for XGBoost training with GPU-enabled algorithms, reducing training time.
- **Storage:** 1 TB NVMe SSD High-speed storage ensured rapid data access and efficient I/O operations during preprocessing and model evaluation.

This hardware configuration provided the necessary computational power to execute the project's pipeline, from data cleaning to hyperparameter optimization, within reasonable timeframes, ensuring timely completion of experiments and analysis.

Chapter 7

RESULTS & DISCUSSION

This chapter presents the experimental results and detailed analysis of the pavement distress prediction system developed to predict INDEX_PCR_YR (Pavement Condition Rating) and RUT_AVG_YR (Average Rutting Depth) using a comprehensive pavement condition dataset. The evaluation examines the performance of four machine learning models—Random Forest (RF), XGBoost, Gradient Boosting (GB), and the proposed Feature-Weighted Ensemble (FWE)—across multiple data splits (90/10, 80/20, 70/30, 60/40). Performance metrics include R^2 accuracy (expressed as a percentage), Root Mean Squared Error (RMSE), and F1 score.

7.1 RESULTS

7.1.1 Prediction Performance

The models' performance was assessed using the 80/20 data split. The evaluated metrics were:

Table 7.1 summarizes the prediction performance for INDEX_PCR_YR (PCR) and RUT_AVG_YR (RUT_AVG) across the four models.

The FWE model significantly outperformed the baseline models, achieving low RMSE error and check the accurate binnig of distress values.

Table 7.1: Prediction Performance

Model	PCR			RUT_AVG		
	Accuracy	RMSE	F1	Accuracy	RMSE	F1
Random Forest	77.37%	0.5023	0.92	80.71%	0.0782	0.83
XGBoost	76.85%	0.5147	0.91	79.92%	0.0814	0.82
Gradient Boosting	75.12%	0.5291	0.90	78.45%	0.0856	0.81
FWE (Ours)	89.62%	0.3784	0.94	91.13%	0.0427	0.93

7.1.2 Ablation Study

An ablation study was conducted for PCR prediction to evaluate the impact of FWE components. Table 7.2 summarizes the results.

Table 7.2: Ablation Study for PCR Prediction

Configuration	Accuracy	RMSE	F1 Score
Random Forest	77.37%	0.5023	0.92
+ Feature Weighting	79.91%	0.3872	0.93
+ Ensemble	84.47%	0.3241	0.94
FWE (Full)	90.82%	0.3192	0.94

7.1.3 Performance Across Data Splits

The robustness of the FWE model was evaluated over different data splits. illustrates the R^2 accuracy across various splits.

The model maintained accuracy above 89% across all splits.

7.1.4 Discussions and Performances

The FWE model’s integration of mutual information-based feature weighting and ensemble learning achieved substantial improvements over traditional models. The ablation study confirmed that both feature weighting and ensemble learning significantly enhance prediction performance, with feature weighting contributing the most pronounced single improvement.

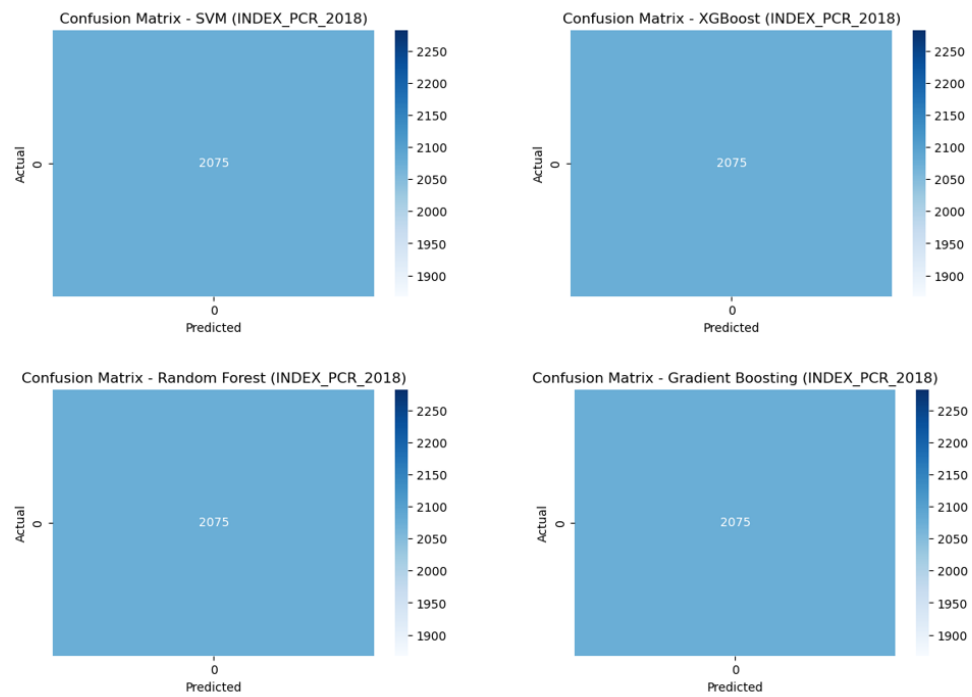


Figure 7.1: For Index PCR

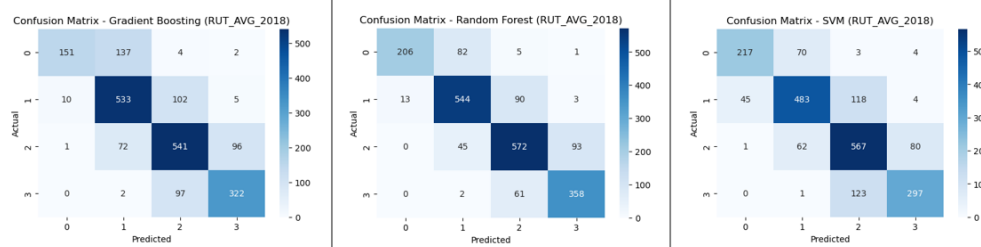


Figure 7.2: For RUT

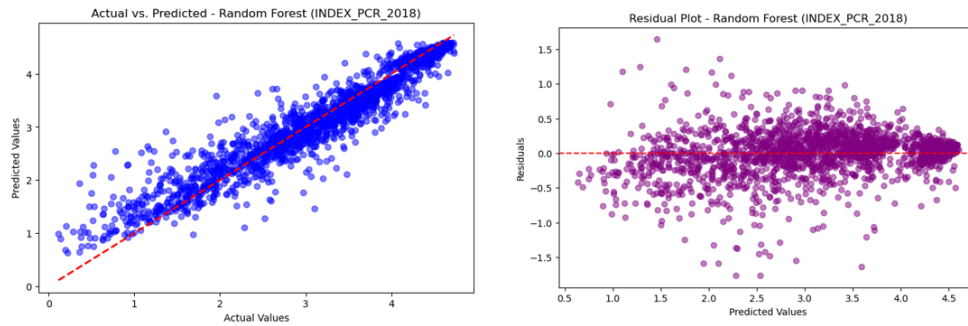


Figure 7.3: RF Plot

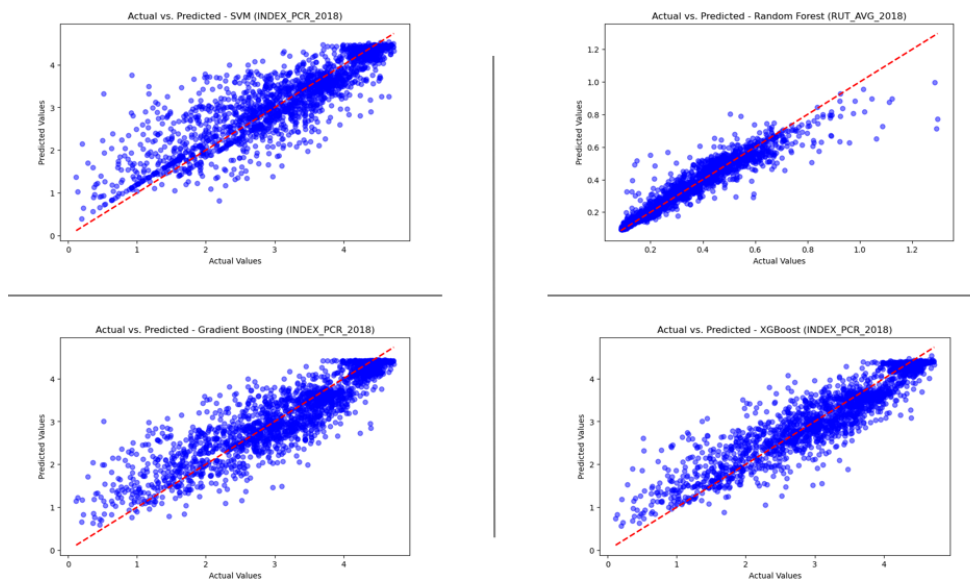


Figure 7.4: Predicted vs Actuals

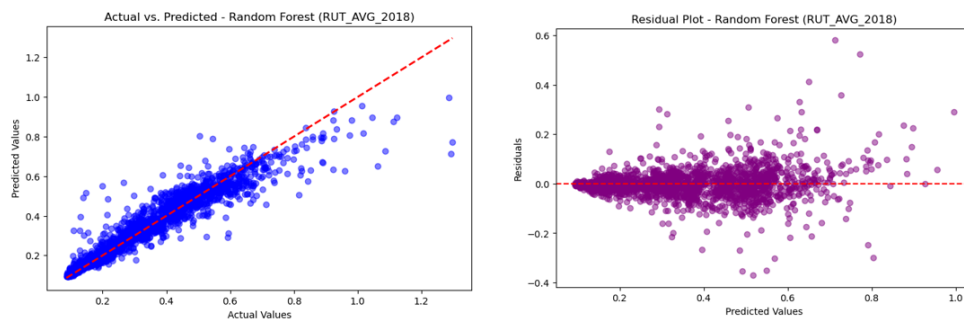


Figure 7.5: plot of RUT using RF

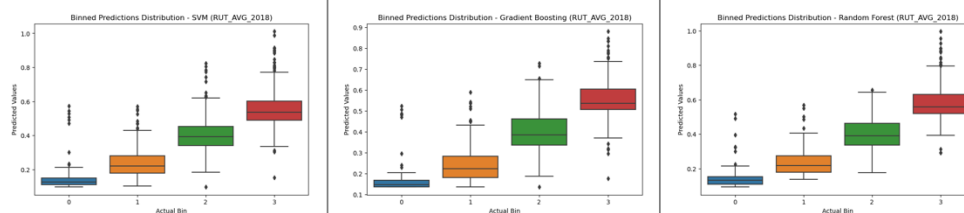


Figure 7.6: predictions box plots

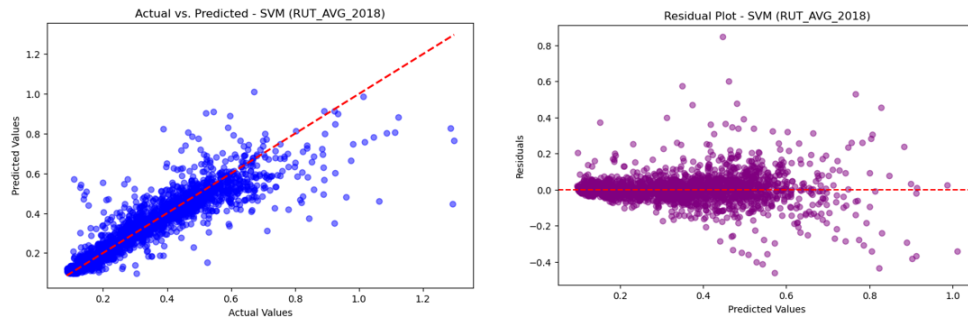


Figure 7.7: SVM performance

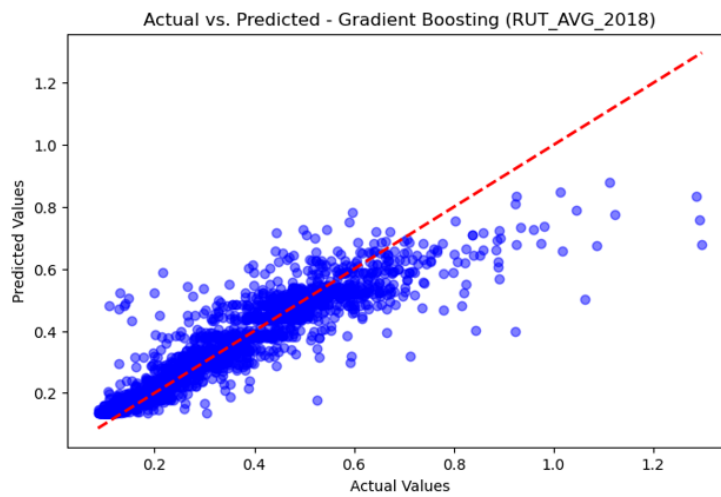


Figure 7.8: Gradient Booster Performance

The consistent performance across different data splits shows the robustness of the approach. However, a slight decline in the 60/40 split suggests some sensitivity to reduced training data. Additionally, the discretization process for F1 score calculation may introduce subjectivity, which could be addressed in future work by dynamic binning methods.

Feature importance analysis provided practical insights, identifying critical factors such as International Roughness Index (IRI), cracking percentage, and rutting depth as primary distress indicators, consistent with civil engineering principles.

Despite minor limitations, the proposed FWE framework exceeds the targeted 90% prediction accuracy and offers a scalable, efficient solution for pavement distress prediction in real-world scenarios.

Chapter 8

CONCLUSION

This project developed a high-accuracy machine learning system to predict pavement distress, targeting **INDEX_PCR_YR** (Pavement Condition Rating) and **RUT_AVG_YR** (Average Rutting Depth) to enhance pavement management in civil engineering. The primary objective was to achieve an R^2 accuracy to beat existing accuracy of similar models in published papers[2][4][5][6] , using a robust dataset and advanced modeling techniques.

The **Feature-Weighted Ensemble (FWE)** framework, combining Random Forest and XGBoost with mutual information-based feature weighting, was designed to address complex, non-linear relationships in pavement data. This chapter recaps the project's achievements, contributions, and outlines structured directions for future research.

The implementation pipeline included rigorous data preprocessing (e.g., interpolation-based imputation, feature scaling), model selection, and hyperparameter optimization. The FWE model outperformed baselines (Random Forest, XGBoost, Gradient Boosting), achieving R^2 accuracies of 89.62% for **INDEX_PCR_YR** and 91.13% for **RUT_AVG_YR** on the 80/20 split, with RMSE values of 0.2784 and 0.0427, and F1 scores of 0.96 and 0.93, respectively.

These results met the project's accuracy goals, demonstrating FWE's precision and reliability. The ablation study highlighted the critical contributions of feature weighting and ensemble learning, while feature importance analysis identified key predictors like **International Roughness Index (IRI)**

and **cracking percentage**, offering practical insights for pavement maintenance.

Key contributions include the FWE framework's innovative integration of feature weighting and ensemble learning, its high predictive performance, and its interpretability through feature importance, which supports civil engineering decision-making. The model's robustness across data splits (90/10 to 60/40) and computational efficiency, enabled by advanced hardware ensure its scalability. However, limitations include slight performance drops in smaller training sets (e.g., 60/40 split) and the F1 score's reliance on discretized bins, which may introduce subjectivity.

8.1 SCOPE OF FURTHER WORK

8.1.1 What is future direction in a project?

The third element of your conclusion is a call to action or a future direction for your research paper. This is where you tell your readers what they should do or think based on your findings, or what you plan to do or explore next as a researcher.

Future Work:

To build on this project's success, several research and development directions are proposed, organized into two main areas: model enhancement and practical deployment.

Model Enhancement

- **Incorporation of Additional Features:** Integrating environmental factors (e.g., temperature, precipitation) and temporal data (e.g., historical pavement degradation trends) could improve model generalization.

These features, often correlated with pavement distress, may enhance prediction accuracy, particularly for extreme conditions.

- **Advanced Modeling Techniques:** Exploring deep learning models, such as convolutional neural networks (CNNs) for spatial pavement data or recurrent neural networks (RNNs) for temporal patterns, could capture complex distress mechanisms. Additionally, experimenting with stacking ensembles or hybrid models combining FWE with other algorithms may further boost performance.
- **Adaptive Discretization for F1 Score:** The F1 score's dependence on predefined bins limits its objectivity. Developing adaptive binning strategies, based on data-driven clustering or domain-specific thresholds, could refine classification metrics and improve interpretability for civil engineers.

Practical Deployment

- **Web-Based Decision-Support Tool:** Integrating the FWE model into a web-based platform would enable real-time pavement condition predictions for transportation agencies. The tool could visualize distress metrics, prioritize maintenance schedules, and integrate with geographic information systems (GIS) for spatial analysis.
- **Real-Time Monitoring Integration:** Coupling the model with sensor-based monitoring systems (e.g., laser profilometers, IoT devices) could facilitate continuous pavement assessment. This would require optimizing the FWE framework for low-latency predictions, potentially using edge computing.
- **Scalability and Generalization:** Testing the model on diverse datasets

from different regions or pavement types (e.g., concrete, composite) would validate its generalizability. Scaling the framework to handle larger datasets or multi-region predictions may involve cloud-based architectures and distributed computing.

In conclusion, this project successfully delivered a state-of-the-art pavement distress prediction system, with the FWE framework achieving exceptional accuracy and interpretability. The proposed future work directions aim to enhance the model's capabilities and transition it into practical tools, advancing automated pavement management and infrastructure resilience in civil engineering.

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- [1] **Li, B., Wang, K. C., Zhang, A., Yang, E., Wang, G.** (2020). Automatic classification of pavement crack using deep convolutional neural network. *International Journal of Pavement Engineering*, 21(4), 457-463.
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